Week 3: User & Group Management (Part 1)

Date: March 23, 2026
Study Time: 5-8 hours total

🎯 GOAL

Establish user/group structure for your infrastructure

📚 STUDY TASKS (1-2 hours)

Reading

- [ ] **"How Linux Works" by Brian Ward**
  - Chapter 7: System Configuration: Users and Groups

Video Learning

- [ ] **LearnLinuxTV** - Search: "User Management Linux"
  - https://youtube.com/@LearnLinuxTV
- [ ] **Professor Messer** - Search: "Linux+ User Management"
  - https://youtube.com/@professormesser

Concepts to Understand

- [ ] /etc/passwd file structure
- [ ] /etc/shadow file (password hashes)
- [ ] /etc/group file structure
- [ ] UID and GID (User ID and Group ID)
- [ ] Home directories (/home/username)
- [ ] Login shells
- [ ] Password policies and aging

🔧 LAB WORK (3-4 hours)

Understanding System Files

- [ ] **Examine /etc/passwd**

```bash
cat /etc/passwd
# Format: username:x:UID:GID:comment:home:shell

# View your own entry
grep $USER /etc/passwd
```

- [ ] **Examine /etc/shadow** (requires sudo)

```bash
sudo cat /etc/shadow
# Contains password hashes and aging info
```

- [ ] **Examine /etc/group**

```bash
cat /etc/group
# Format: groupname:x:GID:members
```

Create Users for Each Tier

- [ ] **Tier 1 admin user**

```bash
sudo useradd -m -s /bin/bash tier1admin
sudo passwd tier1admin

# Verify
id tier1admin
ls -la /home/tier1admin
```

- [ ] **Tier 2 user for Fedora server**

```bash
sudo useradd -m -s /bin/bash fedora_admin
sudo passwd fedora_admin
```

- [ ] **Create service account (no login)**

```bash
sudo useradd -r -s /sbin/nologin gitea_service
# -r = system account, -s = shell (nologin for security)
```

Create Groups

☐ Admin group

```bash
sudo groupadd admin

# Verify
grep admin /etc/group
```

☐ Developers group

```bash
sudo groupadd developers
```

☐ Monitoring group

```bash
sudo groupadd monitoring
```

Add Users to Groups

☐ Add user to admin group

```bash
sudo usermod -aG admin tier1admin
# -aG = append to group (keeps existing groups)

# Verify
groups tier1admin
id tier1admin
```

☐ Add user to multiple groups

```bash
sudo usermod -aG admin,developers,monitoring fedora_admin

# Verify
groups fedora_admin
```

Configure Password Policies

- [ ] **Set password expiry**

```bash
# Password expires in 90 days
sudo chage -M 90 tier1admin

# View password aging info
sudo chage -l tier1admin
```

- [ ] **Force password change on next login**

```bash
sudo chage -d 0 tier1admin
```

- [ ] **Set warning days before expiry**

```bash
sudo chage -W 14 tier1admin
# User warned 14 days before expiry
```

Test User Switching

- [ ] **Switch to another user**

bash

```bash
su - tier1admin
# Enter password

# Check who you are
whoami
id

# Exit back to your user
exit
```

💻 FLASHCARD COMMANDS

Create flashcards for these commands:

1. `useradd`
2. `usermod`
3. `userdel`
4. `groupadd`
5. `groupmod`
6. `passwd`
7. `chage`
8. `id`
9. `groups`
10. `whoami`

📖 EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0)

Relevant Sections:

- 2.5 Given a scenario, manage users and groups
- 3.2 Given a scenario, implement identity management

📝 NOTES & REFLECTIONS

User structure for your infrastructure:

- Tier 1: ______________________
- Tier 2: ______________________
- Tier 3: ______________________
- Tier 4: ______________________

Password policy decisions:

Issues encountered:

Completed: ☐

🔗 QUICK REFERENCE

useradd Options:

- `-m` = create home directory
- `-s` = specify shell
- `-G` = add to supplementary groups
- `-r` = create system account
- `-u` = specify UID

Common UIDs:

- 0 = root
- 1-999 = system accounts
- 1000+ = regular users

Important Files:

- `/etc/passwd` = user accounts
- `/etc/shadow` = password hashes
- `/etc/group` = group definitions
- `/etc/gshadow` = group passwords (rarely used)

Previous Week: Special Permissions
Next Week: sudo Configuration